

Store Sales Forecasting Analysis

Business Analytics Portfolio for HKU-Shenzhen BA Admission



Data Foundation: SQL

构建高效的ETL数据管道，清洗千万级零售交易数据，通过复杂查询挖掘销售趋势与用户行为特征。



Modeling Engine: Python

利用Scikit-learn搭建时间序列预测模型，结合Pandas进行多维数据处理，Matplotlib实现可视化洞察。



Business Impact

精准预测销量波动，优化库存周转与补货策略，有效降低缺货损失与库存积压成本，提升ROI。

Project Background & Goal



01 Business Challenge

Retailers struggle with inefficient inventory control, creating a costly dilemma:

- **Overstock:** Ties up capital, increases storage costs, and risks product obsolescence.
- **Understock:** Leads to lost sales, damaged customer loyalty, and missed revenue.



02 Project Goal

Build a robust, data-driven forecasting model to:

- **Predict:** Accurately forecast future sales trends across stores and categories.
- **Optimize:** Identify key drivers and deliver actionable inventory & marketing strategies.



03 Dataset Source

Utilizing the **Kaggle Store Sales Time-Series Dataset**:

- **Scope:** 54 stores, 3 product categories, spanning 6+ years.
- **Features:** Rich data including promotions, oil prices, holidays, and economic indicators.

“Transforming retail uncertainty into data-driven opportunities to maximize revenue and minimize waste.”

Analytical Workflow



01 Data Preprocessing

Filter invalid data via SQL queries and construct critical business features related to promotions and holidays to ensure data integrity for modeling.



02 Exploratory Analysis

Analyze historical sales trends, identify seasonal fluctuations, and pinpoint peak business periods to uncover underlying data patterns.



03 Feature Engineering

Extract and transform time-based variables (year, month, weekday) and integrate holiday tags to enrich the dataset for robust modeling.



04 Model Building

Establish a Linear Regression baseline and optimize predictive performance using a Random Forest Regressor for higher accuracy.



05 Model Evaluation

Rigorously assess model performance using key metrics (RMSE, MAE) to validate accuracy and ensure reliable predictions for decision-making.



06 Business Recommendation

Translate data-driven insights into actionable business strategies, optimizing resource allocation and enhancing operational efficiency.

Model Result Comparison

01. Linear Regression

A foundational, interpretable model that serves as a reliable baseline for sales forecasting. It establishes clear, linear relationships between variables, offering transparency for initial trend analysis.

Performance: Baseline Benchmark

02. Random Forest

An advanced ensemble learning model that captures complex non-linear patterns and feature interactions. It delivers significantly higher prediction accuracy by aggregating multiple decision trees.

Performance: Optimized & High Accuracy



Key Finding: Critical Growth Drivers

Our analysis identifies **promotional campaigns**, **weekend foot traffic**, and **public holidays** as the top three determinants of sales uplift. These factors create significant spikes in consumer demand, highlighting the need for targeted inventory management and strategic marketing during these high-impact periods.

Business Optimization Suggestions



01 Inventory Preparation

Stock up sufficiently before campaigns and holidays to avoid stockouts. Ensure a stable supply chain to meet customer demand and prevent missed sales opportunities.



02 Peak Staffing & Stock Allocation

Deploy additional staff and prioritize high-demand goods on weekends. This ensures efficient customer service and product availability during peak sales periods.



03 Cost-Efficient Off-Peak Stocking

Reduce stocking quantities on slow weekdays to minimize inventory holding costs and reduce the risk of overstocking. Optimize inventory turnover to free up capital.



04 Strategic Holiday Promotions

Align promotional activities with key holidays and festivals to leverage consumer spending trends. Create targeted campaigns to maximize revenue during high-potential periods.

Summary

This project delivers a complete predictive analytics pipeline, integrating data engineering, machine learning modeling, and retail business interpretation to transform raw data into actionable insights, demonstrating proficiency in quantitative analysis for strategic business decisions.



Data Engineering

Constructed a standardized pipeline for data cleansing, handling missing values, outliers, and normalization to ensure high-quality input for modeling.



ML Modeling

Deployed regression and classification algorithms with cross-validation and hyperparameter tuning to achieve high-precision forecasting of retail trends.



Business Value

Translated model outputs into actionable strategies for inventory optimization, promotional planning, and customer segmentation.

KEY ACHIEVEMENT: Successfully bridged technical modeling with business strategy, delivering a scalable analytics solution that enhances data-driven decision-making and demonstrates core competencies in Business Analytics.



Q&A

Thank You

感谢各位的聆听，欢迎随时提问与交流探讨